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Preface

This document tests workarounds for the non-standard LGR Greek text font encoding for *necessity*, *effects*, and *side-effects*.

The ideal result would be that all Roman numerals came out as expected and the Index would contain two entries.

The new language attributes `local-LGR-fixes` and `no-LGR-fixes` provide a tradeoff: limiting the work-arounds to local scope or skipping them completely helps with the index but breaks some of the Roman numbers. This can help in documents using only small Greek text parts. In the current compilation, Babel is called with the `greek.local-LGR-fixes` and `english` options.

In this document, English text in Greek text parts is not protected against the Greek transliteration (deliberately, to show clearly where Greek is the active language). See the source `*.tex` file for the English original.

1. Problem

- (a) Roman numerals require Latin characters.
- (b) Alphabetic counters use Latin characters, too.
- (c) In document parts using the LGR font encoding, this leads to errors.
 - i. While an “i” is still recognizable after conversion to a small Iota, a “v”, say, becomes an invisible no-break space.
 - ii. The transliteration of “a) b) c) d) e) f) g)” leads to strange results, too.

2. Solution for (a): wrapping roman numbers in `\ensureascii`.

- (a) However, some numbers, e.g. page numbers are auto-generated at “unpredictable” places. Other are used at places “inaccessible” to the author (e.g. the page number of a Greek chapter in the ToC).
 - Re-define counter-formatting functions to make them LGR-proof.
- (b) Moving arguments complicate the matter.
 - i. The proofing must be robust (to some degree) to fix, e.g., a Roman page number of a Greek chapter in the ToC.
 - ii. The proofing must be global to work reliably.

3. Side-effect: the proofing makes Makeindex fail if there are index entries on pages with Roman numbering.

Makeindex expects “pagenumbers” to be simple text strings, without font encoding change or similar. Cf. [Babel issue #170](#).

Προλογος

1. Προβλεμ

(α') Ρομαν νυμεραλς ρεχυιρε Λατιν ζηαφαστερς.

(β') Αλπηαβετις ζουντερς υσε Λατιν ζηαφαστερς, τοο.

(γ') Ιν δοσυμεντ παρτσ υσινγ τηε ΛΓΡ φοντ ενζοδινγ, τηις λεαδς το ερρορς.

i . Ωηιλε αν 'ι' ις στιλλ ρεζογνιζαβλε αφτερ ζονερσιον το α σμαλλ Ιοτα, α ', σαψ, βεζομες αν ινισιβλε νο-βρεακ σπασε.

ii . Α τρανσλιτερατιον οφ 'α) β) ζ) δ) ε) φ) γ)' λεαδς το στρανγε ρεσυλτσ, τοο.

Chapter 1

First Chapter

1.1 Page references

The “preface” is at page [ii](#). The “πρόλογος” is at page [iii](#).

1.2 Page references from a Greek text part

Ο ‘preface’ βρίσκεται στη σελίδα [ii](#). Ο ‘πρόλογος’ βρίσκεται στη σελίδα [iii](#).

- With “local-LGR-fixes”, the page number of the Πρόλογος comes out wrong.
- Authors can fix it by wrapping the `\pageref` function. However, a change of the the default “LGR fix scope” to “local” may break existing documents.

1.3 References to items in an enumeration

See item [1\(c\)ii](#) in an English list. See item [1\(γ\)ii](#) in a Greek list.

1.4 References to items in an enumeration from a Greek text part

Σεε ιτεμ [1\(ς\)ii](#) εν αν Εγγλιση λιστ. Σεε ιτεμ [1\(γ\)ii](#) εν α Γρεεκ λιστ.

- The “alph” counter in the “English” list comes out wrong. (This is also the behaviour in previous versions of babel-greek.)
- Authors must treat references to non-Greek text parts similar to non-Greek words and abbreviations (wrap in `\ensureascii`).

1.5 The Index

The next page should show an Index with 2 entries (see `test-lgr-fixes.idx`).

- With global LGR fixes, no index entry is printed (both are rejected by `makeindex` because of the font encoding change command preceding the page number.
- With “local-LGR-fixes”, only “makeindex-problem” is rejected. (The pagenummer is generated when the `\chapter` command starts the `Προλογος` on a new page. At this point, the active language is Greek and the font encoding LGR.)